

Manoj Kumar Nallavelli

Python Developer and GenAI Engineer

✉ manojnallavelli18@gmail.com ☎ +91 9949667034 📍 Hyderabad, IND

🌐 linkedin.com/in/manojkumar018 🔗 <https://github.com/manoj7034>

PROFILE

Backend-focused Python & GenAI engineer with 3+ years of experience designing and delivering scalable enterprise backend systems, REST APIs, ETL/EDI pipelines, and distributed integration platforms. Skilled in FastAPI, Flask, Kafka, PostgreSQL, Oracle, Linux, and event-driven architectures, with hands-on experience in Generative AI and Machine Learning solutions including RAG pipelines, LangChain, LangGraph, CrewAI, MCP, Vector Databases, and LLM integrations using OpenAI, Anthropic, Gemini, and IBM watsonx. Experienced in building AI-enabled backend services, developing PoCs, integrating intelligent workflows, and delivering production-ready solutions across enterprise environments.

TECHNICAL SKILLS

Programming & Backend

Python, FastAPI, Flask, REST APIs, Microservices Architecture

Database and Caching

PostgreSQL, Oracle, Redis, Vector Databases, Qdrant

EDI & Enterprise Integrations

EDI (X12, EDIFACT), IBM SI/SFG, IBM Control Center, Connect:Direct, ETL Pipelines

Tools & Collaboration

Git, GitHub, Azure Repos, Jenkins, Bitbucket, Apache Superset, ServiceNow, xMatters, Sonar, Agile Development

GenAI & Machine Learning

Generative AI, Machine Learning, LangChain, LangGraph, CrewAI, RAG Pipelines, MCP, AI Orchestration, Semantic Search, Prompt Engineering, OpenAI, Anthropic Claude, Google Gemini, Ollama, Hugging Face, Bedrock, Vertex AI and IBM watsonx

Messaging & Streaming

Apache Kafka, IBM MQ, Logstash

Cloud & DevOps

AWS, GCP, Linux, Docker, VM Deployments, Terraform

AI-Assisted Development Tools

Claude Code, Cursor IDE, GitHub Copilot

PROFESSIONAL EXPERIENCE

Pragma Edge Software Services Pvt Ltd

Python Backend & GenAI Engineer

04/2023

Hyderabad, India

- Designed and Developed scalable backend services and REST APIs using **Python, FastAPI, and Flask** for enterprise monitoring, workflow automation, and distributed integration platforms.
- Built ETL/EDI processing pipelines and event correlation systems integrating databases, Kafka, MQ, Webhooks, and APIs across enterprise environments.
- Implemented SLA monitoring, intelligent alerting, and operational reporting integrations using **ServiceNow, xMatters, and Apache Superset**.
- Integrated AI-enabled backend services with **IBM watsonx**, exposing secure APIs for intelligent recommendations, conversational workflows, and automated reprocessing.
- Built and integrated **RAG-based AI workflows** using LangChain, Vector Databases, and LLM APIs to enable context-aware recommendations and intelligent automation.
- Developed AI orchestration services using **LangChain, LangGraph, and CrewAI** for multi-step reasoning, workflow execution, and conversational operations.
- Integrated LLM providers including **OpenAI, Anthropic Claude, and Google Gemini** into backend systems using secure REST APIs, semantic search, and structured prompt workflows.
- Implemented prompt engineering, response validation, and AI workflow optimization to improve reliability and contextual accuracy of AI-generated outputs.
- Participated in technical solutioning, architecture discussions, and PoC development for backend and AI-enabled enterprise workflows.

- Collaborated with cross-functional and client-facing teams to gather requirements, review implementations, and deliver scalable production-ready solutions.
- Contributed to deployment workflows, integration pipelines, and backend optimization initiatives across distributed enterprise platforms.
- Worked directly with **enterprise clients** on production support, deployments, feature enhancements, and technical demonstrations across backend and AI-enabled platforms.

PROJECTS

IANN FileGPS – AI-Powered Enterprise File & Transaction Monitoring Platform

- Enterprise-grade platform providing real-time, end-to-end visibility into file transfers, EDI transactions, and business workflows across distributed enterprise systems through scalable monitoring, event correlation, SLA tracking, and intelligent alerting workflows.
- Built backend services, event processing pipelines, and monitoring integrations using databases, messaging systems, logs, APIs, Kafka, and MQ to capture, enrich, aggregate, and contextualize enterprise file and transaction events.
- Enhanced the platform with AI/ML and NLP-driven capabilities using IBM watsonx, LangChain, and RAG-based workflows to support intelligent recommendations, anomaly detection, predictive error analysis, conversational operations, and automated file reprocessing.
- Implemented AI-enabled monitoring for file anomalies, system health anomalies, authentication failures, and SLA deviations to proactively detect operational risks and unexpected behaviors across enterprise environments.
- Developed chatbot and AI-assisted operational workflows using LangChain and RAG pipelines to enable contextual querying, alert analysis, workflow recommendations, and intelligent issue resolution across projects, partners, and transaction data.
- Built customizable alerting and reporting workflows supporting dashboard, email, and notification-based operational visibility for enterprise monitoring and compliance management.

PragmaQ – AI-Powered IT Service Management Platform

- Built AI-powered backend services for an enterprise ITSM platform using Python/FastAPI microservices, LangChain, LangGraph, and CrewAI orchestration for multi-agent workflows and LLM-driven automation.
- Designed and implemented RAG pipelines using TF-IDF, FAISS, Qdrant vector search, and LangChain retrieval chains to surface context-aware KB articles, SOPs, and historical tickets.
- Developed AI-driven workflows for ticket classification, incident correlation, Root Cause Analysis, and SOP recommendation using DistilBERT, all-MiniLM-L6-v2, and Google Gemini.
- Implemented MCP-compatible AI services exposing ticketing, workflow, and knowledge-base operations as callable tools for agent-driven automation and intelligent workflow execution.
- Built semantic retrieval and AI response generation systems using vector similarity search, prompt engineering, structured LLM output parsing, and conversational AI workflows.
- Developed scalable AI execution pipelines using Redis-backed scheduling, RabbitMQ messaging, and multi-worker FastAPI services for distributed AI workflow processing.

ACHIEVEMENTS & RECOGNITION

Extra Miler Award

Awarded for consistently going above and beyond to achieve excellence.

Employee of the Month

Awarded for consistent delivery, backend ownership, and resolution of critical production issues.

Customer Champion Award

Customer Champion Award for exceptional client engagement and ownership of enterprise implementations.

EDUCATION

Masters in Science

Osmania University

03/2021 – 10/2022

Hyderabad, India